

CSE 881: Data Mining

Lecture 6: Rule-based Classifier

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Rule-Based Classifier

- Classify records by using a collection of “if...then...” rules
- Rule: $(Condition) \rightarrow y$
 - where
 - ◆ *Condition* is a conjunctions of attributes
 - ◆ *y* is the class label
 - *LHS*: rule antecedent or condition
 - *RHS*: rule consequent
 - Examples of classification rules:
 - ◆ $(Blood\ Type=Warm) \wedge (Lay\ Eggs=Yes) \rightarrow Birds$
 - ◆ $(Taxable\ Income < 50K) \wedge (Refund=Yes) \rightarrow Evade=No$

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Rule-based Classifier (Example)

Name	Body Temperature	Skin Cover	Gives Birth	Aquatic Creature	Aerial Creature	Has Legs	Hibernates	Class Label
human	warm-blooded	hair	yes	no	no	yes	no	Mammals
python	cold-blooded	scales	no	no	no	no	yes	Reptiles
salmon	cold-blooded	scales	no	yes	no	no	no	Fishes
whale	warm-blooded	hair	yes	yes	no	no	no	Mammals
frog	cold-blooded	none	no	semi	no	yes	yes	Amphibians
komodo dragon	cold-blooded	scales	no	no	no	yes	no	Reptiles
bat	warm-blooded	hair	yes	no	yes	yes	yes	Mammals
pigeon	warm-blooded	feathers	no	no	yes	yes	no	Birds
cat	warm-blooded	fur	yes	no	no	yes	no	Mammals
guppy	cold-blooded	scales	yes	yes	no	no	no	Fishes
alligator	cold-blooded	scales	no	semi	no	yes	no	Reptiles
penguin	warm-blooded	feathers	no	semi	no	yes	no	Birds
porcupine	warm-blooded	quills	yes	no	no	yes	yes	Mammals
eel	cold-blooded	scales	no	yes	no	no	no	Fishes
salamander	cold-blooded	none	no	semi	no	yes	yes	Amphibians

$r_1: (\text{Gives Birth} = \text{no}) \wedge (\text{Aerial Creature} = \text{yes}) \rightarrow \text{Birds}$
 $r_2: (\text{Gives Birth} = \text{no}) \wedge (\text{Aquatic Creature} = \text{yes}) \rightarrow \text{Fishes}$
 $r_3: (\text{Gives Birth} = \text{yes}) \wedge (\text{Body Temperature} = \text{warm-blooded}) \rightarrow \text{Mammals}$
 $r_4: (\text{Gives Birth} = \text{no}) \wedge (\text{Aerial Creature} = \text{no}) \rightarrow \text{Reptiles}$
 $r_5: (\text{Aquatic Creature} = \text{semi}) \rightarrow \text{Amphibians}$

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Application of Rule-Based Classifier

- A rule r **covers** an instance x if the attributes of the instance satisfy the condition of the rule

$r_1: (\text{Gives Birth} = \text{no}) \wedge (\text{Aerial Creature} = \text{yes}) \rightarrow \text{Birds}$
 $r_2: (\text{Gives Birth} = \text{no}) \wedge (\text{Aquatic Creature} = \text{yes}) \rightarrow \text{Fishes}$
 $r_3: (\text{Gives Birth} = \text{yes}) \wedge (\text{Body Temperature} = \text{warm-blooded}) \rightarrow \text{Mammals}$
 $r_4: (\text{Gives Birth} = \text{no}) \wedge (\text{Aerial Creature} = \text{no}) \rightarrow \text{Reptiles}$
 $r_5: (\text{Aquatic Creature} = \text{semi}) \rightarrow \text{Amphibians}$

Name	Body Temperature	Skin Cover	Gives Birth	Aquatic Creature	Aerial Creature	Has Legs	Hibernates
hawk	warm-blooded	feather	no	no	yes	yes	no
grizzly bear	warm-blooded	fur	yes	no	no	yes	yes

The rule r_1 covers a hawk $\Rightarrow$ Bird

The rule r_3 covers the grizzly bear $\Rightarrow$ Mammal

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Rule Coverage and Accuracy

- Coverage of a rule:
 - Fraction of records that satisfy the antecedent of a rule
- Accuracy of a rule:
 - Fraction of records that satisfy both the antecedent and consequent of a rule

Tid	Refund	Marital Status	Taxable Income	Class
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

(Status=Single) → No

Coverage = 40%, Accuracy = 50%

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How does Rule-based Classifier Work?

r_1 : (Gives Birth = no) $\wedge$ (Aerial Creature = yes) $\rightarrow$ Birds
 r_2 : (Gives Birth = no) $\wedge$ (Aquatic Creature = yes) $\rightarrow$ Fishes
 r_3 : (Gives Birth = yes) $\wedge$ (Body Temperature = warm-blooded) $\rightarrow$ Mammals
 r_4 : (Gives Birth = no) $\wedge$ (Aerial Creature = no) $\rightarrow$ Reptiles
 r_5 : (Aquatic Creature = semi) $\rightarrow$ Amphibians

Name	Body Temperature	Skin Cover	Gives Birth	Aquatic Creature	Aerial Creature	Has Legs	Hibernates
lemur	warm-blooded	fur	yes	no	no	yes	yes
turtle	cold-blooded	scales	no	semi	no	yes	no
dogfish shark	cold-blooded	scales	yes	yes	no	no	no

A lemur triggers rule r_3 , so it is classified as a mammal

A turtle triggers both r_4 and r_5

A dogfish shark triggers none of the rules

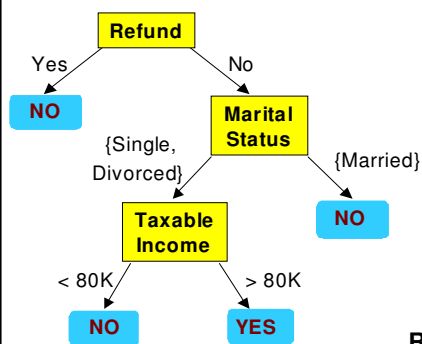
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Characteristics of Rule-Based Classifier

- Mutually exclusive rules
 - Classifier contains mutually exclusive rules if the rules are independent of each other
 - Every record is covered by at most one rule
- Exhaustive rules
 - Classifier has exhaustive coverage if it accounts for every possible combination of attribute values
 - Each record is covered by at least one rule

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From Decision Trees To Rules



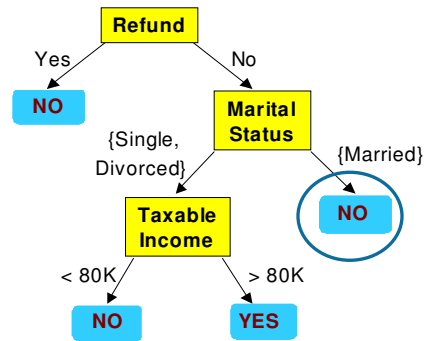
Classification Rules

(Refund=Yes) ==> No
(Refund=No, Marital Status={Single,Divorced}, Taxable Income<80K) ==> No
(Refund=No, Marital Status={Single,Divorced}, Taxable Income>80K) ==> Yes
(Refund=No, Marital Status={Married}) ==> No

Rules are mutually exclusive and exhaustive
Rule set contains as much information as the tree

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Rules Can Be Simplified



Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Initial Rule: $(\text{Refund}=\text{No}) \wedge (\text{Status}=\text{Married}) \rightarrow \text{No}$

Simplified Rule: $(\text{Status}=\text{Married}) \rightarrow \text{No}$

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Effect of Rule Simplification

- Rules are no longer mutually exclusive
 - A record may trigger more than one rule
 - Solution?
 - Ordered rule set
 - Unordered rule set – use voting schemes
- Rules are no longer exhaustive
 - A record may not trigger any rules
 - Solution?
 - Use a default class

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Ordered Rule Set

- Rules are rank ordered according to their priority
 - An ordered rule set is known as a decision list
- When a test record is presented to the classifier
 - It is assigned to the class label of the highest ranked rule it has triggered
 - If none of the rules fired, it is assigned to the default class

r_1 : (Gives Birth = no) $\wedge$ (Aerial Creature = yes) $\rightarrow$ Birds
 r_2 : (Gives Birth = no) $\wedge$ (Aquatic Creature = yes) $\rightarrow$ Fishes
 r_3 : (Gives Birth = yes) $\wedge$ (Body Temperature = warm-blooded) $\rightarrow$ Mammals
 r_4 : (Gives Birth = no) $\wedge$ (Aerial Creature = no) $\rightarrow$ Reptiles
 r_5 : (Aquatic Creature = semi) $\rightarrow$ Amphibians

Name	Body Temperature	Skin Cover	Gives Birth	Aquatic Creature	Aerial Creature	Has Legs	Hibernates
turtle	cold-blooded	scales	no	semi	no	yes	no

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Rule Ordering Schemes

- Rule-based ordering
 - Individual rules are ranked based on their quality/priority
- Class-based ordering
 - Rules that belong to the same class appear together

Rule-Based Ordering

(Skin Cover=feathers, Aerial Creature=yes) $\Rightarrow$ Birds
 (Body temperature=warm-blooded, Gives Birth=yes) $\Rightarrow$ Mammals
 (Body temperature=warm-blooded, Gives Birth=no) $\Rightarrow$ Birds
 (Aquatic Creature=semi) $\Rightarrow$ Amphibians
 (Skin Cover=scales, Aquatic Creature=no) $\Rightarrow$ Reptiles
 (Skin Cover=scales, Aquatic Creature=yes) $\Rightarrow$ Fishes
 (Skin Cover=none) $\Rightarrow$ Amphibians

Class-Based Ordering

(Skin Cover=feathers, Aerial Creature=yes) $\Rightarrow$ Birds
 (Body temperature=warm-blooded, Gives Birth=no) $\Rightarrow$ Birds
 (Body temperature=warm-blooded, Gives Birth=yes) $\Rightarrow$ Mammals
 (Aquatic Creature=semi) $\Rightarrow$ Amphibians
 (Skin Cover=none) $\Rightarrow$ Amphibians
 (Skin Cover=scales, Aquatic Creature=no) $\Rightarrow$ Reptiles
 (Skin Cover=scales, Aquatic Creature=yes) $\Rightarrow$ Fishes

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Building Classification Rules

- Direct Method:
 - ◆ Extract rules directly from data
 - ◆ e.g.: RIPPER, CN2, 1R, and AQ
- Indirect Method:
 - ◆ Extract rules from other classification models (e.g. decision trees, neural networks, SVM, etc).
 - ◆ e.g: C4.5rules

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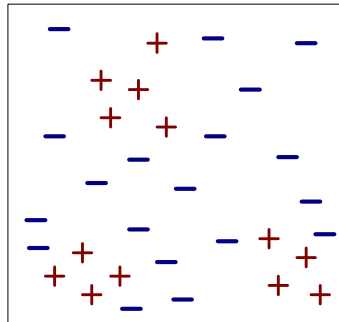
Direct Method: Sequential Covering

Algorithm 1.1 Sequential covering algorithm.

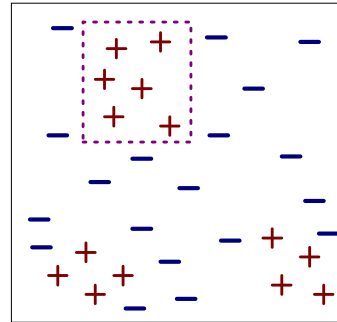
- 1: Let E be the training records and A be the set of attribute-value pairs, $\{(A_j, v_j)\}$.
 - 2: Let Y_o be an ordered set of classes $\{y_1, y_2, \dots, y_k\}$.
 - 3: Let $R = \{ \}$ be the initial rule list.
 - 4: **for** each class $y \in Y_o - \{y_k\}$ **do**
 - 5: **while** stopping condition is not met **do**
 - 6: $r \leftarrow \text{Learn-One-Rule}(E, A, y)$.
 - 7: Remove training records from E that are covered by r .
 - 8: Add r to the bottom of the rule list: $R \longrightarrow R \vee r$.
 - 9: **end while**
 - 10: **end for**
 - 11: Insert the default rule, $\{ \} \longrightarrow y_k$, to the bottom of the rule list R .
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Example of Sequential Covering



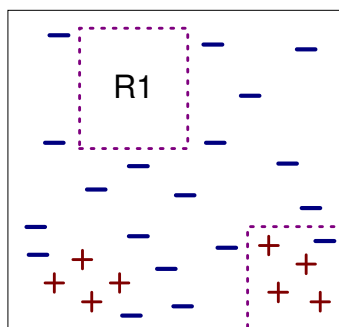
(i) Original Data



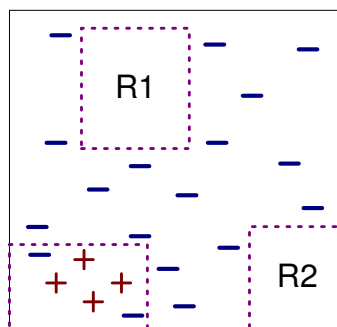
(ii) Step 1

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Example of Sequential Covering...



(iii) Step 2



(iv) Step 3

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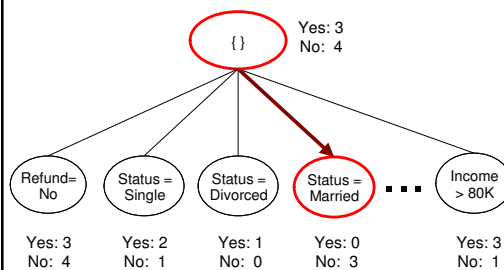
Aspects of Sequential Covering

- Rule Growing
 - Rule evaluation
- Instance Elimination
- Stopping Criterion
- Rule Pruning

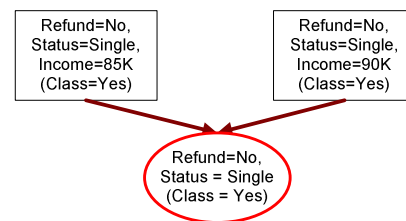
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Rule Growing

- Two common strategies



(a) General-to-specific



(b) Specific-to-general

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Rule Evaluation

- Evaluation metric determines which conjunct should be added during rule growing

- Accuracy = $\frac{n_c}{n}$

- Laplace = $\frac{n_c + 1}{n + k}$

- M-estimate = $\frac{n_c + kp}{n + k}$

n : Number of instances covered by rule

n_c : Number of instances covered by rule

k : Number of classes

p : Prior probability

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Rule Growing (Examples)

- CN2 Algorithm:

- Start from an empty conjunct: {}
 - Add conjuncts that minimizes the entropy measure: {A}, {A,B}, ...
 - Determine the rule consequent by taking majority class of instances covered by the rule

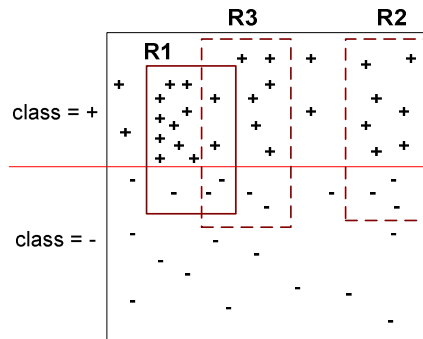
- RIPPER Algorithm:

- Start from an empty rule: {} => class
 - Add conjuncts that maximizes FOIL's information gain measure:
 - ♦ R0: {} => class (initial rule)
 - ♦ R1: {A} => class (rule after adding conjunct)
 - ♦ Gain(R0, R1) = $t [\log(p1/(p1+n1)) - \log(p0/(p0 + n0))]$
 - ♦ where t : number of positive instances covered by both R0 and R1
 - p0: number of positive instances covered by R0
 - n0: number of negative instances covered by R0
 - p1: number of positive instances covered by R1
 - n1: number of negative instances covered by R1

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Instance Elimination

- Why do we need to eliminate instances?
 - Otherwise, the next rule is identical to previous rule
- Why do we remove positive instances?
 - Ensure that the next rule is different
- Why do we remove negative instances?
 - Prevent underestimating accuracy of rule
 - Compare rules R2 and R3 in the diagram



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Stopping Criterion and Rule Pruning

- Examples of stopping criterion:
 - If rule does not improve significantly after adding conjunct
 - If rule starts covering examples from another class
- Rule Pruning
 - Similar to post-pruning of decision trees
 - Example: using validation set (reduced error pruning)
 - ◆ Remove one of the conjuncts in the rule
 - ◆ Compare error rate on validation set before and after pruning
 - ◆ If error improves, prune the conjunct

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Summary of Direct Method

- Initial rule set is empty
- Repeat
 - Grow a single rule
 - Remove Instances covered by the rule
 - Prune the rule (if necessary)
 - Add rule to the current rule set

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Direct Method: RIPPER

- For 2-class problem, choose one of the classes as positive class, and the other as negative class
 - Learn the rules for positive class
 - Use negative class as default
- For multi-class problem
 - Order the classes according to increasing class prevalence (fraction of instances that belong to a particular class)
 - Learn the rule set for smallest class first, treat the rest as negative class
 - Repeat with next smallest class as positive class

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Direct Method: RIPPER

- Rule growing:
 - Start from an empty rule: $\{\} \rightarrow +$
 - Add conjuncts as long as they improve FOIL's information gain
 - Stop when rule no longer covers negative examples
 - Prune the rule immediately using incremental reduced error pruning
 - Measure for pruning: $v = (p-n)/(p+n)$
 - ♦ p : number of positive examples covered by the rule in the validation set
 - ♦ n : number of negative examples covered by the rule in the validation set
 - Pruning method: delete any final sequence of conditions that maximizes v

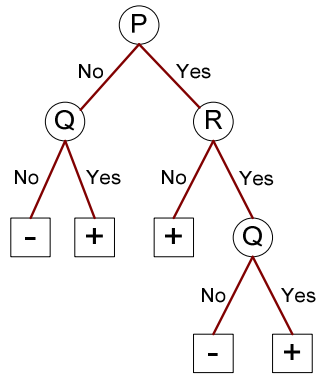
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Direct Method: RIPPER

- Building a Rule Set:
 - Use sequential covering algorithm
 - ♦ Grow a rule to cover the current set of positive examples
 - ♦ Eliminate both positive and negative examples covered by the rule
 - Each time a rule is added to the rule set, compute the new description length
 - ♦ stop adding new rules when the new description length is d bits longer than the smallest description length obtained so far

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Indirect Methods



Rule Set

r1: (P=No,Q=No) ==> -
r2: (P=No,Q=Yes) ==> +
r3: (P=Yes,R=No) ==> +
r4: (P=Yes,R=Yes,Q=No) ==> -
r5: (P=Yes,R=Yes,Q=Yes) ==> +

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Indirect Method: C4.5rules

- Extract rules for every path from root to leaf nodes
- For each rule, $r: A \rightarrow y$,
 - consider alternative rule $r': A' \rightarrow y$ where A' is obtained by removing one of the conjuncts in A
 - Compare the pessimistic error rate for r against all r 's
 - ◆ Prune if one of the r 's has lower pessimistic error rate
 - Repeat until pessimistic error rate can no longer be improved

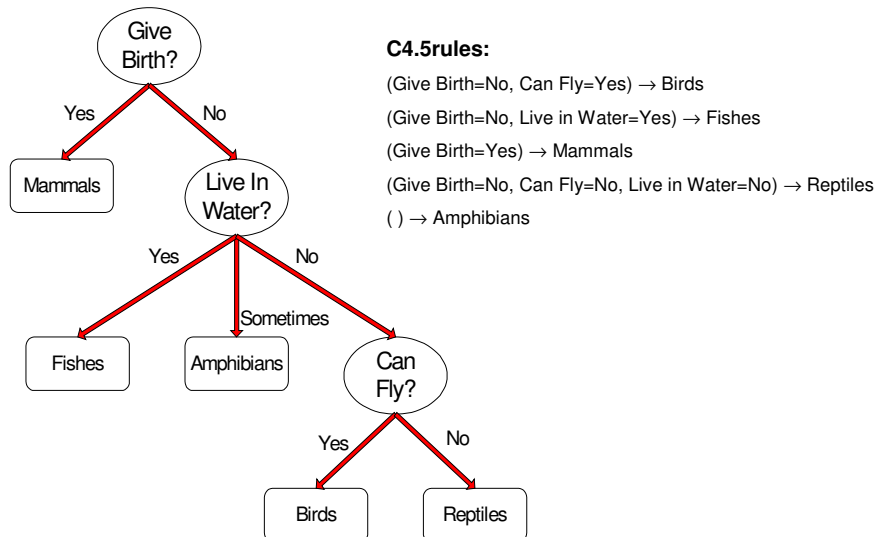
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Indirect Method: C4.5rules

- Use class-based ordering
 - Rules that predict the same class are grouped together into the same subset
 - Compute total description length for each class
 - Classes are ordered in increasing order of their total description length

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Example



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Characteristics of Rule-Based Classifiers

- As highly expressive as decision trees
- Easy to interpret
- Easy to generate
- Can classify new instances rapidly
- Performance comparable to decision trees